

Lucas Saraiva Teixeira

DOCTORAL RESEARCHER IN ELECTRICAL ENGINEERING

 Campinas, SP, Brazil  lucas.sarvtex@gmail.com  +55 (31) 98470-2188  <https://brenergies.org/>  [sarvtex](#)
 [lucassaraivateixeira](#)  0009-0005-8102-8834

Electrical Engineer researcher studying Physics-Informed Machine Learning (PIML) as applied to Lithium Battery health prognosis and health management, including simulation, estimation, and prediction.

EDUCATION

Ph.D. in Electrical Engineering

State University of Campinas (UNICAMP)

2025 – present

Campinas, SP, Brazil

Thesis: *Physics-Informed Machine Learning for Lithium Battery Health Management and Prognosis (WIP)*.

Advisor: Prof. Hudson Zanin.

M.Sc. in Electrical Engineering

Federal University of Minas Gerais (UFMG)

2022 – 2024

Belo Horizonte, MG, Brazil

Dissertation: *Modelling Lithium-Ion Battery Degradation for Second Life Stationary Applications*.

Advisor: Prof. Sidelmo Magalhães Silva

B.Sc. in Electrical Engineering

Federal University of Minas Gerais (UFMG)

2016 – 2021

Belo Horizonte, MG, Brazil

RESEARCH EXPERIENCE

Doctoral Researcher

BRnergies (UNICAMP)

2025 – present

Campinas, SP, Brazil

- Project in Development. Applying Physics-Informed Machine Learning principles to Battery “Prognosis and Health Management”.
- Use PyBaMM and COMSOL for generating Battery simulated data (parameter estimation and synthetic training data).
- Use Pytorch for building Deep-Learning models.

Master Student Researcher

Tesla Lab (UFMG)

2022 – 2024

Belo Horizonte, MG, Brazil

- Research battery degradation simulation with electrochemical models via PyBaMM.

Academic Exchange Program

Brno University of Technology

2020

Brno Czechia

- Testing Intelligent Electronic Devices effects of electrical substation safety. Involve both traditional simulations with PSCAD and Hardware-in-the-Loop simulations with RSCAD.

SELECTED PUBLICATIONS

- 1 G. B. Sanches, C. A. Rufino Jr., **L. S. Teixeira**, H.-G. Schweiger, and H. G. Zanin. Single-cycle prognostics of lithium-ion batteries using statistical feature engineering and extremely randomized tree ensembles. *Results in Engineering*, Vol **32**, 111218 (2026).
- 2 **L. S. Teixeira** and S. M. da Silva. Simulating electric-vehicle battery degradation with physics based battery model. *Congresso Brasileiro de Automática – CBA* **4**(1) (2024).

SKILLS

Programming Python, Matlab

Methods Deep Learning, Physics-Informed Machine Learning, Battery Simulation

Tools PyTorch, Git, $\LaTeX$, PyBAMM

LANGUAGES

English	Fluent
Portuguese	Native
Mandarim	Novice